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1. Introduction

This document contains the 2018 Management Report of the administration of the Corporation Center for Excellence in Marine Sciences - CEMarin. The Report is divided into six sections. The first section details the academic and scientific management; the second reports on the management of funding sources; the third includes a description of the activities related to communications management; the fourth refers to strategic management; the fifth summarizes financial and tax management; and the sixth, presents the main updates in the area of human resources.

2. Academic and scientific management

The academic component of CEMarin addresses various pressing problems covered by marine sciences, and allows students to become familiar with methodologies and innovative research topics at a global level. Its main objective is to offer comprehensive training to postgraduate students in marine sciences from any part of the world, while enabling the interaction of young and renowned international researchers.

Among the academic activities carried out during 2018, the International Conference on Marine Sciences, the CEMarin Cátedra and the virtual course “Deep-Sea Exploration – CEMarin” stand out.

Colombia 2018: International Conference on Marine Sciences

From November 19th to 23rd, 2018, the “Colombia 2018: International Conference on Marine Sciences - Towards a Sustainable Ocean” event was held in Medellín and jointly organized by the Corporation Center of Excellence in Marine Sciences (CEMarin), the Inter-Institutional Doctorate in Marine Sciences (IDMS) and the Symposium on Water Waves (LatWaves). The conference was held at the Universidad Nacional de

Colombia, Medellín Campus, in the Alejandro López Auditorium of the Faculty of Mines, at Robledo campus.

The main objective of the conference was to bring together researchers, practitioners and educators to exchange and share their experiences in answering fundamental environmental and socio-economical questions related to marine ecosystems in Latin America. Furthermore, it aimed to provide an interdisciplinary forum for discussing environmental change and its impacts on ecosystems and society, the use of marine and coastal ecosystems, as well as technological advances. The conference focused on the following themes:

1. Marine environmental change: From species responses to environmental modeling
 2. Marine resources: Sustainable use in a changing world.
 3. Ocean and society: The social dimension of ecosystem services.
 4. Oceanography: Understanding the physical processes of atmosphere-ocean interactions
 5. Water waves: How the ocean affects life
- The event gathered together more than 200 scientists, decision-makers, politicians, educators and business representatives from 24 countries. As the theme of the conference was “Towards a sustainable ocean” participants discussed emerging issues and challenges related to the protection and sustainable use of marine resources both



in Colombia and in the rest of the world, stemming from different fields of research in global change, biology, fishery sciences, oceanography and socio-economic and political science.

During the opening ceremony, former CEMarin Executive Director, Professor Jairo Zapata Martínez, introduced the other speakers - Verónica Botero Fernández, Dean of the Faculty of Mines of the Universidad Nacional de Colombia., followed by Eduardo Rojas Pineda, Director of Research Promotion at the Colombian Administrative Department of Science, Technology, and Innovation (Colciencias).

The opening event also included words from Ana María González, delegate of the Department of Marine, Coastal and Aquatic Affairs of the Ministry of Environment and Sustainable Development, Colombia. Finally, the H.E. Peter Ptassek, German Ambassador to Colombia, closed the opening session by celebrating the model of international scientific cooperation between the German Academic Exchange Service (DAAD) and the CEMarin Corporation, a model they have held for ten years.

Opening



Jairo Zapata Martínez
Executive director



Ana María González Delgadillo
Department of Marine, Coastal and Aquatic Affairs Delegate



Verónica Catalina Botero Fernández.
Faculty of Mines Dean
Universidad Nacional de Colombia, Medellín headquarters



Peter Ptassek
Ambassador of Germany in Colombia



Eduardo Rojas Pineda
Research Promotion Director
Colciencias

Colombia 2018:
**International Conference
on Marine Science**
Towards a sustainable ocean

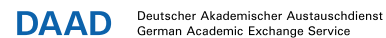


Figure 1. The speakers of the inaugural session of the International Conference on Marine Sciences.



Figure 2. International keynote speakers of the International Conference on Marine Sciences.

The special programming of the conference included sixteen outstanding plenaries in marine sciences; some of the keynote speakers were: Maren Ziegler, Alec Torres Freyermuth, Patricia Miloslavich, Javier López Lara, Nelson Violante Carvalho, Rodrigo Figueroa, Mathias Wolff and Takuji Waseda. In total, there were eighty-one talks by representatives from different countries and the event ended with the exhibition of 36 posters on November 22nd and 23rd.

At the end of the Conference four awards were given to the best scientific talks; the winners were: Myriam Vargas, Jessica Reichert, Paula Andrea Zapata, Rafael Ricardo Torres. The award for the best poster presentation went to Luis Alberto Guerra.

Thanks to the support of sponsors Lat-Waves and CEMarin, financed by DAAD, and through the coordination of the interinstitutional doctorate program of Marine Sciences, 49 travel grants were given to participants and speakers.

Conference Statistics

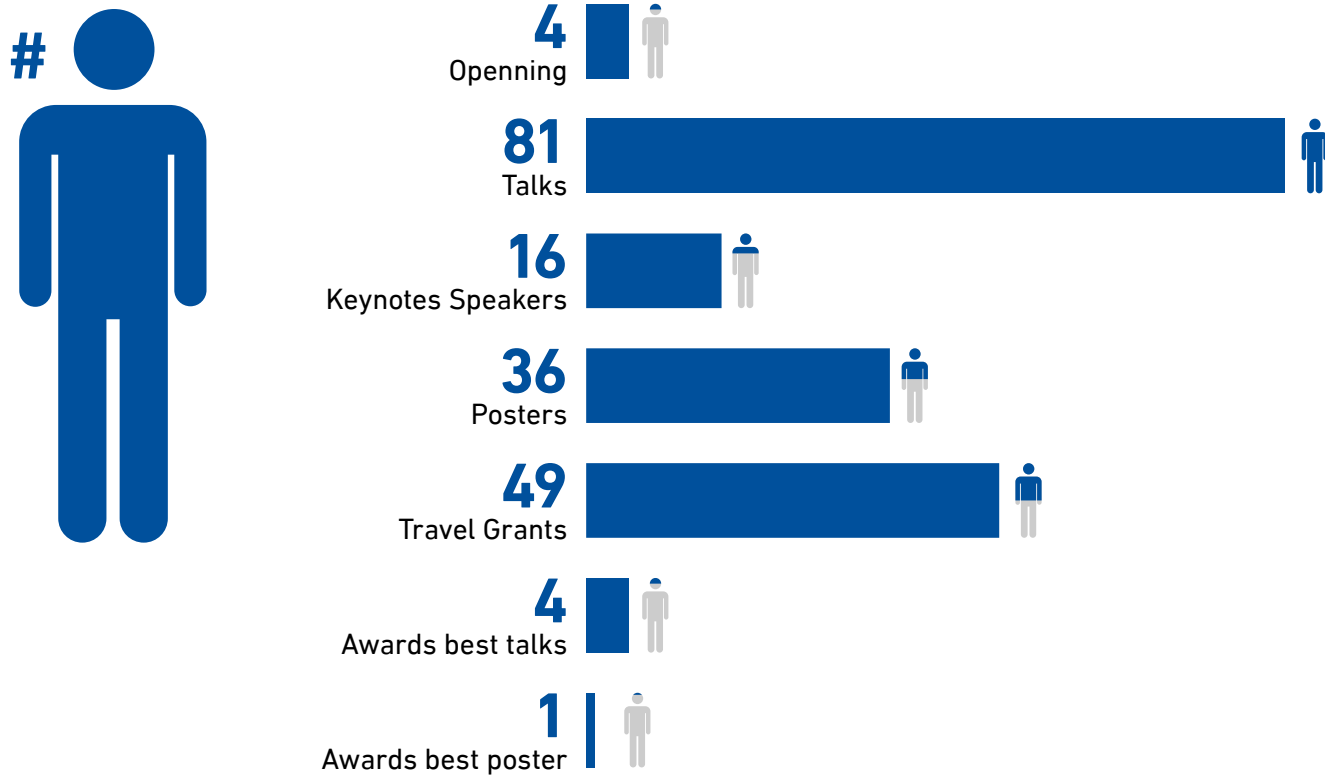


Figure 3. General statistics of the International Conference on Marine Sciences.

Assistance statistics

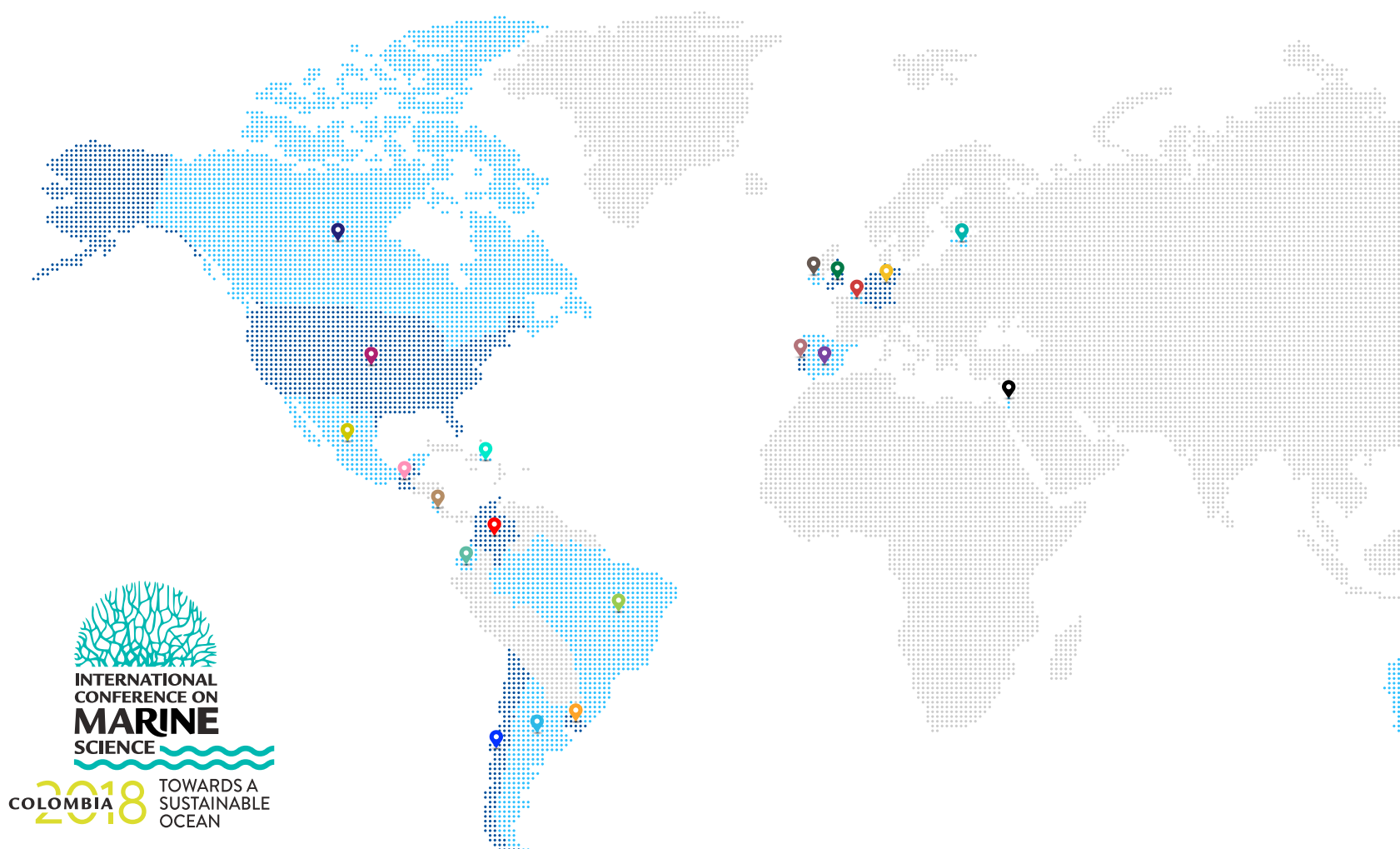
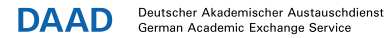


Figure 4. Assistance statistics for the International Conference on Marine Sciences.



Number of attendees per country

24
Countries

TOTAL
Attendees
222

Argentina 1	Colombia 1	Germany 1	New Zealand 1
Australia 3	Costa Rica 2	Guatemala 1	Portugal 1
Belgium 1	Dominican republic 2	Ireland 1	South Korea 1
Brazil 5	Ecuador 5	Israel 1	Spain 6
Canada 1	England 1	Japan 1	United States 1
Chile 1	Estonia 1	Mexico 1	Uruguay 1

155



Countries with highest
number of attendees

12



Colombia

11



Germany

7



United States

Mexico

CEMarin Cathedra

The keynote lectures held jointly by the member universities of CEMarin Corporation are called CEMarin Cátedra. The short-courses or lectures are transmitted online in real-time and are aimed at a broad audience; participants do not necessarily need to be specialized in the field. Two short-courses were part of the Cátedra in 2018.

The first took place during the first semester of the year: Among the subjects offered by the Graduate school of the Department of Biological Sciences of the Universidad de Los Andes, the virtual course “Deep-Sea Exploration - CEMarin” was given by the CEMarin researcher Juan Armando Sánchez.

As part of the University's contribution to the Corporation, the course was opened free or charge to any student linked to the universities associated with CEMarin. Seven students benefited from this opportunity: Kara Engelhardt, Denise Klinkenbuss and Kirsten Kuhbier from the Justus-Liebig University; Juan Sebastián Giraldo, Adriana Sarmiento and Diana Vergara from Universidad de los Andes; and Sandra Montaña from Universidad Nacional de Colombia.

 Universidad de los Andes	Departamento de Ciencias Biológicas		CRN30504	CBIO4529
	Deep-Sea Exploration CEMarin		Version No.	1
			Date	15-01-2018
			Virtual	



Title: ‘Deep-Sea Exploration-CEMarin’¹

Code: CRN30504, CBIO4529 **Credits:** 4

Academic calendar: 2018-1 (March through May)

Place: On-line hosted at <https://sicuaplus.uniandes.edu.co/webapps/login/>

Juan Armando Sánchez, Ph.D., Professor and course coordinator

Departamento de Ciencias Biológicas, Universidad de los Andes Email: juansanc@uniandes.edu.co

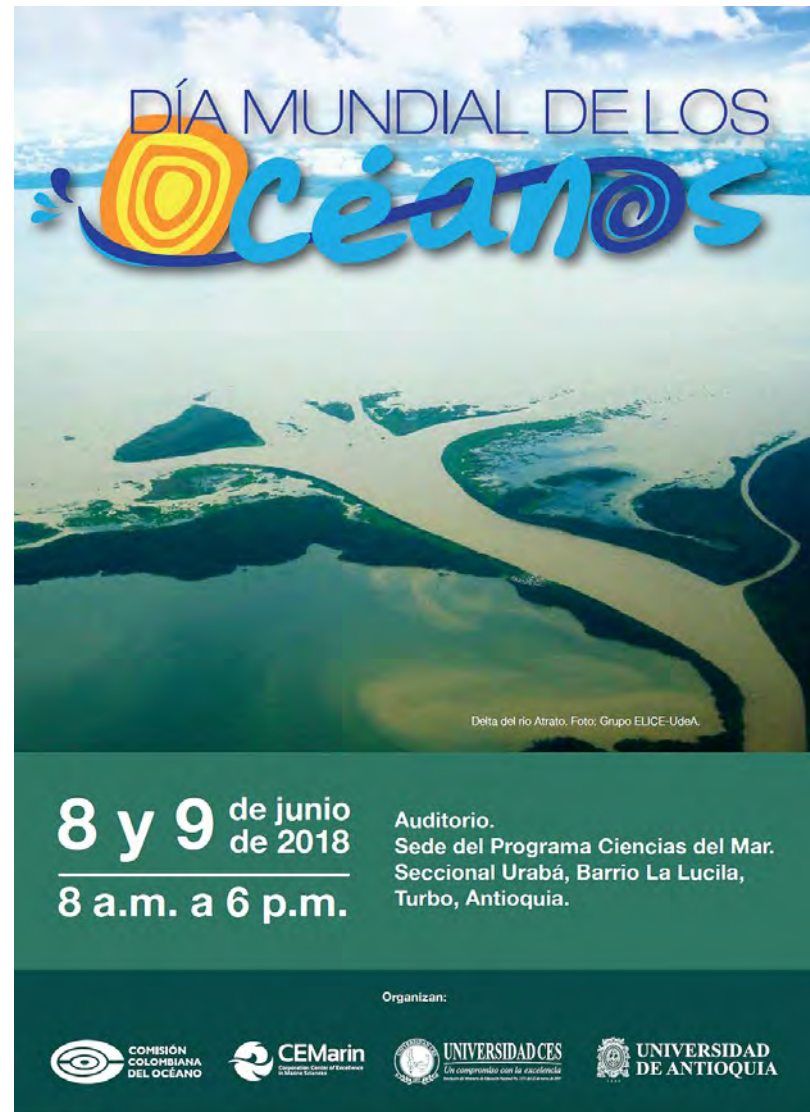
“You can only inspire when you give people a new way of looking at the world in which they live.” Robert Ballard, pioneer deep-sea explorer

Figure 5. Deep-Sea Exploration – CEMarin course offer.

During the second semester of the year the Universidad de Antioquia, the Universidad CES, the Colombian Commission of the Ocean and the CEMarin Corporation organized a second short-course: "VIII Academic Meeting: Let's Clean Up Our Oceans" that took place in commemoration of World Oceans Day.

National and international experts participated in the event and they presented research focused especially on the issue of ocean pollution caused by plastic. The meeting was held in Medellín at the Universidad CES and the Universidad de Antioquia from June 6th to 9th, and at the Headquarters of Marine Sciences, Universidad de Antioquia, in the municipality of Turbo, on June 8th and 9th.

During the meeting, former Executive Director of CEMarin Professor Jairo Zapata Martínez, was the moderator of the discussion "What can we do for the oceans? The marine ecosystems from our responsibility". and the CEMarin Researcher Néstor Hernando Campos participated as a speaker in the session "What is contaminating the Colombian Caribbean? Changes in time".



DÍA MUNDIAL DE LOS Océanos

Delta del río Atrato. Foto: Grupo EUCE-UdeA.

8 y 9 de junio de 2018
8 a.m. a 6 p.m.

**Auditorio.
Sede del Programa Ciencias del Mar.
Seccional Urabá, Barrio La Lucila,
Turbo, Antioquia.**

Organizan:

COMISIÓN COLOMBIANA DEL OCEANO

CEMarin

UNIVERSIDAD CES

UNIVERSIDAD DE ANTIOQUIA

Figure 6. VIII Academic Meeting: Let's clean up our oceans program.



Calls.

Current calls

Call No. 3: CEMarin Early Stage Researchers (ESR)

Call No. 3: CEMarin Early Stage Researchers (ESR)

This call, which was opened on September 10th, 2015, and is permanently open, is addressed to doctoral students belonging to the partner institutions of the Corporation. The objective is to link early stage researchers (ESR) as official members of CEMarin and provide an opportunity for them to share their research projects with other students and researchers of marine sciences. By being part of the Corporation, ESR can apply for the CEMarin calls, access financial support offered to help develop their thesis projects, and attend courses, workshops, field trips, etc.

During 2018, five (5) students were linked to the Corporation as CEMarin ESR; currently there are a total of thirty-three (33) doctoral students enrolled under this category.

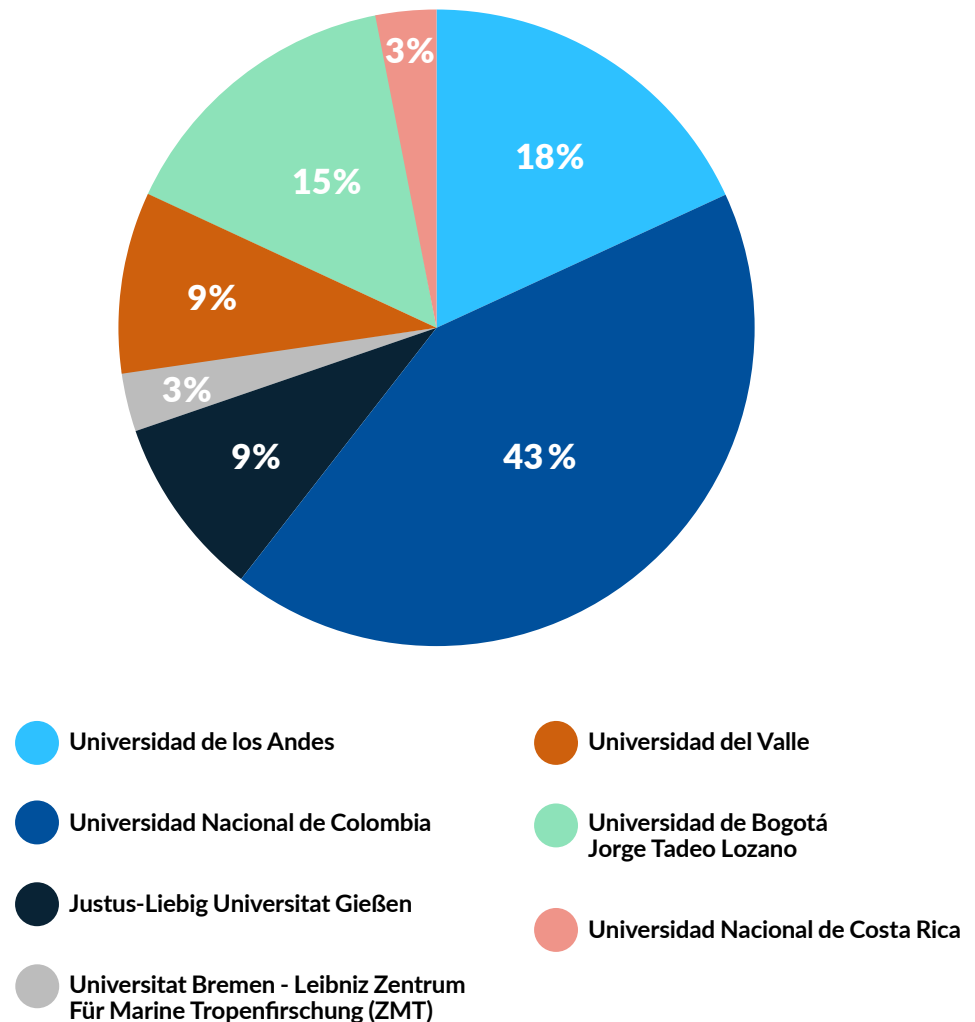


Figure 7. Percentage of ESR per institution.

Call No. 12: Financial support for doctoral thesis projects

This call was opened on July 4th, 2018, and will remain open until all the financial resources available, a total of \$ 160 million COP, are allocated to CEMarin ESR. The objective is to support ESR financially for the development of their doctoral thesis projects in marine sciences and related subjects.

During 2018 three proposals were received and submitted to the Scientific Committee for evaluation and all three received their endorsement for financing. To date, the three students who benefited, legalized their contracts with CEMarin and began to use the resources.

CEMarin ESR	Doctoral Program	Institution	Name of Thesis Project	Amount approved for financing \$COP
Estefanía Isaza Toro	Inter-institutional Doctorate in Marine Sciences	Universidad del Valle	Prediction and seasonality of potential habitats for Dolphinfin (Coryphaena hippurus) and Tuna (Scombridae) and their relationship with environmental changes in the Tropical Eastern Pacific Ocean	5.000.000
Milena Benavides Serrato	Doctorate in Sciences - Biology, Marine Biology line	Universidad Nacional de Colombia	Connectivity between natural population of sea urchin Echinometra lucunter lucunter (Echinodermata: Echinoidea: Echinometridae) throughout Caribbean region	9.408.569
Dalia Barragán Barrera	Doctorate in Biological Sciences	Universidad de los Andes	Combining stable isotope and mercury analyses to assess coastal habits of a vulnerable bottlenose dolphin population in the Panamanian Caribbean	14.200.000

Table 1. Winning projects of the Call No. 12.

Call No. 13: Seed Grants

This call, which opened on June 27th, 2018, will remain open until all financial resources available, a total of \$ 120 million COP, are allocated to CEMarin Researchers. The objective is to support CEMarin Researchers with seed grants; researchers must be in the process of preparing interdisciplinary research proposals that address novel questions in marine sciences, and which will subsequently be submitted to national and international funding agency calls. The resulting projects are expected to achieve financing greater than USD 50.000 per year.

During 2018, two proposals were received for evaluation by the Scientific Committee. Both received their endorsement to be financed and, to date, the CEMarin researchers who benefited legalized their contracts with the Corporation and began to use the resources.

CEMarin Researchers	Institutions	Name of the Project	Amount approved for financing
Camilo García & Mónica Puyana	Universidad Nacional de Colombia & Universidad de Bogotá Jorge Tadeo Lozano	Assessment of marine biodiversity in the Colombian Caribbean sea in times of climate change: Recovering the past, understanding the present and projecting the future	12.000.000
Jaime Cantera & Andrés Osorio	Universidad del Valle & Universidad Nacional de Colombia	Shallow-water habitat mapping in remote tropical coasts: evaluating the usefulness of lightweight unmanned aerial vehicles for conservation and monitoring	12.000.000

Table 2. Winning projects of the Call No. 13.

Former calls

Call No. 5: Financial support for doctoral thesis projects

This call, which opened on September 10th, 2015, and remained open until January 15th, 2016, was addressed to CEMarin ESR. CEMarin committed a total of \$ 246.891.700 COP (two hundred and forty-six million, eight hundred and ninety-one thousand, seven hundred Colombian pesos) in order to financially support beneficiaries with the development of their doctoral thesis projects in marine sciences and related topics.

During 2018, after receiving the respective reports on financial execution and progress in academic commitments, CEMarin verified the information and disbursed the total approved amount to six of the nine CEMarin ESR who benefited from the call. Furthermore, one ESR received the second of the 3 agreed disbursements. Thus, at the end of the year \$ 218.945.850 COP of the total resources assigned to financing the beneficiary students had been executed.

CEMarin ESR	Doctoral program	Institution	Disbursements	Date of last disbursement	Total disbursed
Diana Milena Bustos Montes	Doctorate in Sciences - Biology, Marine Biology line	Universidad Nacional de Colombia	3 of 3	November 9th, 2018	30.000.000
Ángel Germán Franco Franco	Doctorate in Natural Sciences	Justus-Liebig University	3 of 3	November 30th, 2018	30.000.000
Ernesto Joaquín Acosta Ortiz	Doctorate in Natural Sciences	Justus-Liebig University	3 of 3	October 5th, 2018	30.000.000
Luis Alberto Guerra Vargas	Doctorate in Marine Sciences	Universidad Nacional de Colombia	3 of 3	January 19th, 2018	30.000.000
Pilar Adriana Herrón Pérez	Doctorate in Natural Sciences	University of Bremen	3 of 3	February 21st, 2018	21.000.000
Iván Fernando Calixto Botía	Doctorate in Biological Sciences	Universidad de los Andes	3 of 3	December 19th, 2018	30.000.000
Elena Quintanilla Alcaide	Doctorate in Biological Sciences	Universidad de los Andes	2 of 3	June 8th, 2018	20.000.000
Juan Guillermo García Garay	Doctorate in Biological Sciences	Universidad de Bogotá Jorge Tadeo Lozano	2 of 3	October 6th, 2017	20.000.000

Table 3. Resources executed in the Call No. 5.

Call No. 8: Financial support for doctoral thesis projects

This call, which opened on May 11th, 2016, and remained open until October 20th of the same year was addressed to CEMarin ESR. CEMarin committed a total of \$ 185.740.000 COP (one hundred eighty-five million, seven hundred forty thousand Colombian pesos) in order to financially support beneficiaries in the development of their doctoral thesis projects in marine sciences and related topics.

During 2018, after receiving the respective reports on financial execution and progress in academic commitments, CEMarin verified the information and disbursed the total approved amount to four (4) of the eight (8) CEMarin ESR who benefited from the call. Furthermore, two (2) ESR received the second of the 3 agreed disbursements. Thus, at the end of the year \$ 152 million COP (one hundred fifty-two million Colombian pesos) of the total resources assigned to financing the beneficiary students had been executed.

CEMarin ESR	Doctoral program	Institution	Disbursements	Date of last disbursement	Total disbursed
Juan Sebastián Celis	Doctorate in Natural Sciences	Justus-Liebig University	3 of 3	June 1st, 2018	21.000.000
Natalia Beatriz Comba González	Doctorate in Sciences - Biology, Marine Biology line	Universidad Nacional de Colombia	3 of 3	November 9th, 2018	15.000.000
Juan Carlos Mejía Rentería	Inter-institutional Doctorate in Marine Sciences	Universidad del Valle	3 of 3	June 1st, 2018	21.000.000
Adriana Sarmiento Segura	Doctorate in Biological Sciences	Universidad de los Andes	3 of 3	April 13th, 2018	30.000.000
Gilberto Junior Orozco Berdugo	Inter-institutional Doctorate in Marine Sciences	Universidad Nacional de Colombia	2 of 3	May 11th, 2018	20.000.000
Juan Carlos Narváez Barandica	Doctorate in Biology, Marine Biology line	Universidad Nacional de Colombia	2 of 3	December 4th, 2018	20.000.000
Estefania Isaza Toro	Inter-institutional Doctorate in Marine Sciences	Universidad del Valle	2 of 3	September 11th, 2017	21.000.000
Argiro de Jesús Ramírez Aristizabal	Doctorate in Natural Sciences for Development – DOCINADE	Universidad Nacional de Costa Rica	1 of 2	July 12th, 2017	4.000.000

Table 4. Resources executed in the Call No. 8.

Call No. 14: Financial support for doctoral students within a research project

This call, which opened on December 13th, 2018, and remained open until February 18th, 2019, was addressed to CEMarin Researchers. The objective of the call is to financially support Colombian and German CEMarin ESR who are linked to a research project led by a CEMarin Researcher and who also want to carry out their research outside their home country, whether in Colombia or Germany.

The proposals presented were evaluated by the Scientific Committee who determined which proposals will receive financing.

During 2019 the legalization of the respective contracts with the CEMarin researchers will be carried out and the execution of the financial resources will begin.

CEMarin researcher	Institution	Name of the Project
Thomas Wilke	Justus-Liebig University	Physiological response of stony corals to microplastic exposure
Jorge Maldonado	Universidad de los Andes	Evaluating allocation rules for managing marine fisheries in developing countries
Andrés Osorio	Universidad Nacional de Colombia	Hydrodynamics of coral reefs: Dispersion patterns and response under global change conditions
José Ernesto Mancera	Universidad Nacional de Colombia	Adaptability through cross-sowing of seedlings of three mangrove species (<i>Rhizophora mangle</i> , <i>Laguncularia racemosa</i> , <i>Avicennia germinans</i>) in environments with different salinity regimes on the island of San Andrés (Colombia) as a technical input for the restoration of Neotropical mangrove forests
Adriana Santos	Universidad Nacional de Colombia	Relationships between coral reef complexity and ecosystem services at Caribbean oceanic islands, Seaflower Biosphere Reserve, Colombia
José Julián Tavera	Universidad del Valle	Tempo and mode of diversification in sister species of marine fishes separated by the Isthmus of Panama

Table 5. Winning projects of the Call No. 14.

A close-up photograph of a seal's head above the surface of the ocean. The seal has dark fur, large eyes, and prominent whiskers. The water is a deep blue with gentle ripples. The seal is looking directly at the camera.

3. Funding sources management

In the last two years, CEMarin Corporation has sought to expand its sources of income beyond the contributions made by its members. To fulfil this purpose, CEMarin hired a professional, Mauricio Romero, to take charge of the management of scientific projects. During 2018, the Scientific project manager designed a strategy for mapping of potential funders for research proposals that included potential international funding sources and actions to strengthening the links with strategic national and regional partners. The strategy had four components:

1. Development of databases and bulletins:

Keyword-specific protocol searches for all thematic areas of CEMarin were developed. Some of the primary databases for international funding sources (e.g., Pivot, Research Professional and Scopus) and national funding sources (e.g., Presidential Agency for International Cooperation and SECOP II) were used. Subsequently, these

funding opportunities were analyzed and a database with 131 potential sources of funding was built based on relevant information. As a result, ten bulletins with 70 funding sources were sent to all CEMarin researchers (see Fig. 8) and, in addition, a list with funding sources for undergraduate and graduate students was sent to several researchers.

1	
TYPE	GRANT
INSTITUTION_NAME	KR Foundation
FUNDING OPPORTUNITY TITLE	Funding
CLOSING_DATE_(D/M/Y)	January 2019 (Forecast)
AMOUNT	\$300.000-700.000 USD or 5,000,000 - 2,000,000 DKK
URL	http://krfnd.org/apply-for-funding/
Description	KR Foundation addresses root causes of climate change and environmental degradation, and safeguard the planetary boundaries.
	The mission of KR Foundation is to help provide answers to, stimulate mind shifts about, and encourage action on, the long-term challenges faced by current and future generations living on a planet with finite resources, fragile ecosystems, and climate change. Read more about our mission here.
	As a philanthropic foundation, KR Foundation typically engages in areas where neither governments nor markets actors seem to have incentives to act. Not restricted by short-term considerations and profit concerns, KR Foundation engages in non-profit activities with long-term impact on our climate and environment.
	KR Foundation priorities communication and advocacy activities within our program areas but will also consider proposals in other areas with a high potential for impact that meet our general criteria for funding. Please contact the Foundation prior to submitting a proposal that does not match the Foundation's program areas. Please note that KR Foundation does not fund local and community-based activities unless they are truly scalable and have a clear international perspective. Proposals focusing on local development activities will not be considered.
General criteria for funding KR Foundation address root causes of climate change and environmental degradation, through supporting activities that:	

Figure 8. Bulletin sent to CEMarin researchers with funding opportunities.

2. Submitting existing projects to new calls and potential donors:

Existing or developing research projects were matched with new calls and donors. For example:

a. For international funding sources: the project “Renewable energies in the insular Caribbean region”, led by Professor Andrés Osorio (Grupo Oceánicos, Universidad Nacional de Colombia, Medellín headquarters), was submitted to the call for proposals from

Euroclimate+ seeking funding of 1.5 million Euros (see Fig. 9). The Corporation for the Sustainable Development of the San Andrés Archipelago, Providencia and Santa Catalina (CORALINA) and the Bluerise Renewable Energy Company also participated in this proposal. The proposal did not advance from the prequalification phase.

b. For local funding sources, the project “Assessment of marine biodiversity in the Colombian Caribbean Sea in times of climate

change” was submitted to the Maritime Training and Instruction Center (CIMAR). The proposal was written by professors Camilo García (Universidad Nacional de Colombia, Bogotá headquarters) and Mónica Puyana (Universidad de Bogotá Jorge Tadeo Lozano); The project obtained an initial grant through the CEMarin call No. 13 and is currently being reformulated.



EUROCLIMA +

Componente Eficiencia Energética

Convocatoria de Proyectos

Anexo 2 – Ficha de Preformulación de Proyecto

Desarrollo de capacidades locales e innovativas en energías renovables para la promoción de eficiencia energética en la región Caribe Insular

Presentado por

Corporación para el Desarrollo Sostenible del Archipiélago de San Andrés,

Providencia y Santa Catalina, Colombia



Pequeños científicos del océano

En la región Caribe colombiana, la próxima generación de ciudadanos resolverá sus problemas socio-ambientales con competencias basadas en ciencia, tecnología, ingeniería y matemáticas (STEM). Sin embargo, en las escuelas del Caribe existe un rezago en competencias STEM que requiere de un programa de educación de alta calidad en estas áreas. El programa Pequeños Científicos del Océano actualizará a los docentes de educación básica y media en nuevos métodos de enseñanza de competencias STEM. Se espera que por medio de unidades de aprendizaje, módulos didácticos y una constante autoevaluación, el proyecto promueva el pensamiento crítico, la indagación científica para la resolución de problemas complejos y promueva a los jóvenes en campos como la programación, las matemáticas y la física. Las áreas STEM son la puerta de entrada para comprender la biodiversidad, el océano, el clima y en últimas, el uso sostenible de los recursos marino-costeros. La pedagogía de STEM se ha probado exitosamente en 100 colegios de Latinoamérica y Colombia.

Problemática a abordar. El deterioro ambiental es un problema complejo y resolverlo efectivamente representa un inmenso desafío técnico y educativo. Aunque las instituciones educativas oficiales del Caribe han mejorado significativamente su cobertura y disminuido la deserción escolar, poseen un rezago en la calidad de educación con valores inferiores a la media nacional [1]. Las experiencias mundiales indican que el camino para mejorar la calidad educativa requiere de intervenciones desde la primera infancia hasta el fin de la educación secundaria. El Caribe necesita una nueva y probada estrategia para mejorar la educación básica y media que incluya la formación docente de alta calidad en áreas STEM.

Es prioritario enfrentar los desafíos de la sociedad actual como la salud, seguridad nacional, nuevas energías, ambiente y uso sostenible de los recursos.

Objetivos

1. Formar 50 docentes en STEM para la sostenibilidad ambiental.
2. Entregar módulos didácticos.
3. Acompañar el diseño e implementación de evaluaciones de estudiantes y docentes.



Tiempo de ejecución y costo
12 meses
85 millones de pesos

Entidades participantes

CEMARIN: www.cemarin.org
Pequeños Científicos: www.pequenoscientificos.org

Lugar de ejecución y población involucrada
Municipio de Arbolotes.
Profesores y niños de primaria y secundaria

Figure 9. Examples of projects submitted to international calls. Left: Renewable energy project led by Professor Andrés Osorio (UNal, Medellín). Right: Small ocean scientists project led by CEMarin and the NGO Pequeños Científicos.

3. The construction of new research projects with local and regional partners: The proposal “Small ocean scientists” was built with the NGO Pequeños Científicos (see Fig. 9). This proposal was submitted to the Anadarko Colombia Oil Company with a value of \$ 120 million COP (one hundred and twenty million Colombian pesos); so far no response has been received about the proposal. On the other hand, the project summary “Grant program for research projects in marine sciences” was written-up and submitted to the Packard Foundation for a grant of \$ 120.000 USD (one hundred and twenty thousand US dollars). The Foundation responded that the proposal was outside its area of interest. The project was also submitted to the Ford Foundation with no response to date.

4. Publishing CEMarin Corporation calls in our partner universities. Three presentations were made to advertise the current calls of the Corporation at the Universidad de Antioquia, at the Universidad de Bogotá Jorge Tadeo Lozano and at the Universidad Nacional de Colombia, and the bulletin of international calls to which CEMarin researchers can postulate their projects was presented.



4. Communications management

During 2018, CEMarin created and executed a digital communication strategy through which the disclosure of the Corporation's activities was achieved through its social networks and the publication of content on its website. In addition, the promotion of the International Conference on Marine Sciences was strengthened due to the strategic alliances established with other institutions.

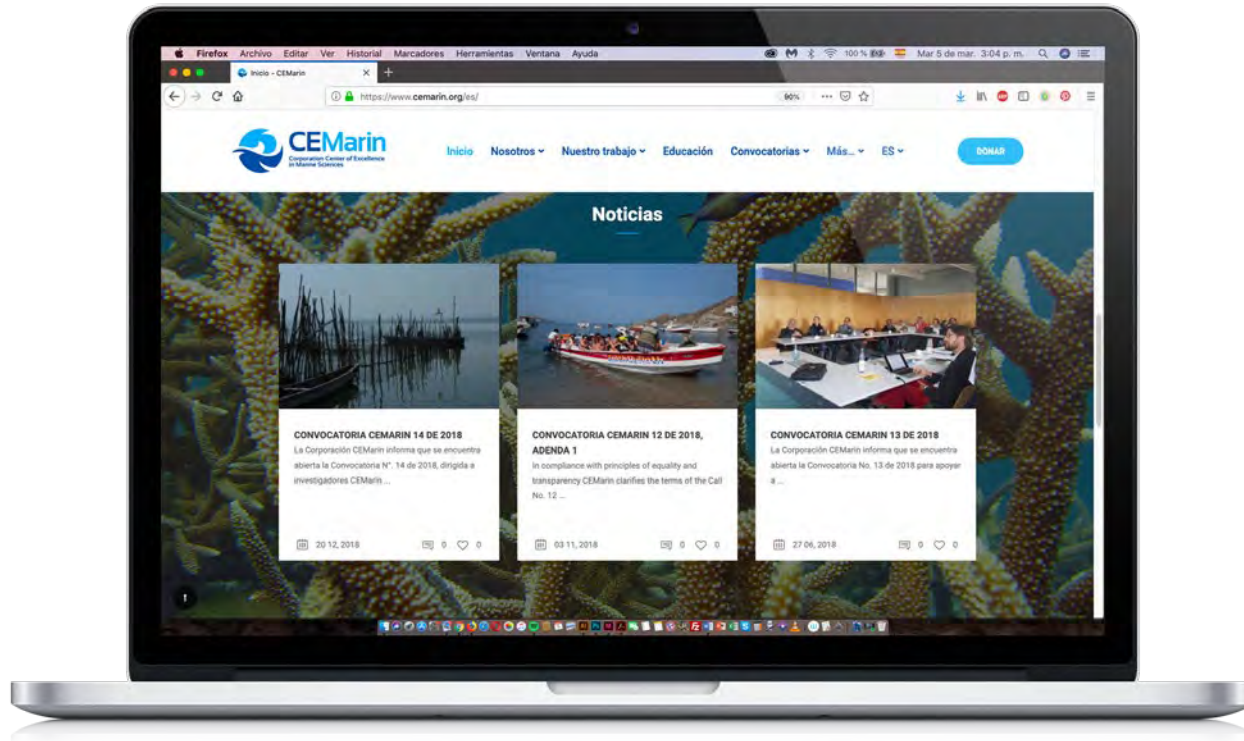


Figure 10. Design and distribution of the Home - News home page of the website.

Website

During the first semester, modifications were made to the website architecture and new roles were assigned to content editors. The new design improved the ability to search for information because it is visually more accessible for the user.

Regarding to the website content, the main objective of the changes were to share CEMarin's events such as the "VIII Academic meeting: Clean up our oceans" in the framework of the World Day of the Oceans celebrations and the virtual course "Deep-Sea Exploration

- CEMarin" and the International Conference on Marine Sciences. These events were also disseminated by email marketing. Likewise, information about calls No. 3, 12, 13, and 14 were shared via website and new content (articles) were created for the website, for example - "Ciencia y conciencia para cuidar los océanos", "La dura tarea de limpiar los océanos", "Germany and Colombia: A joint investigation of the ocean", "Tarena, oceanographic expedition in the Gulf of Urabá" and "Pedro Pablo Vallejo, el químico del mar".

Social networks

At the beginning of the year CEMarin had a small presence on social network platforms. On Facebook, the Corporation's page had only 136 followers, and CEMarin did not have a Twitter account. A strategy was designed to increase the Corporation's digital followers that consisted of creating content that was published periodically on Facebook and Twitter. The content that was created focused on CEMarin's calls and, mainly, the International Conference on Marine Sciences. The follower growth over the year is shown in the graphic below.

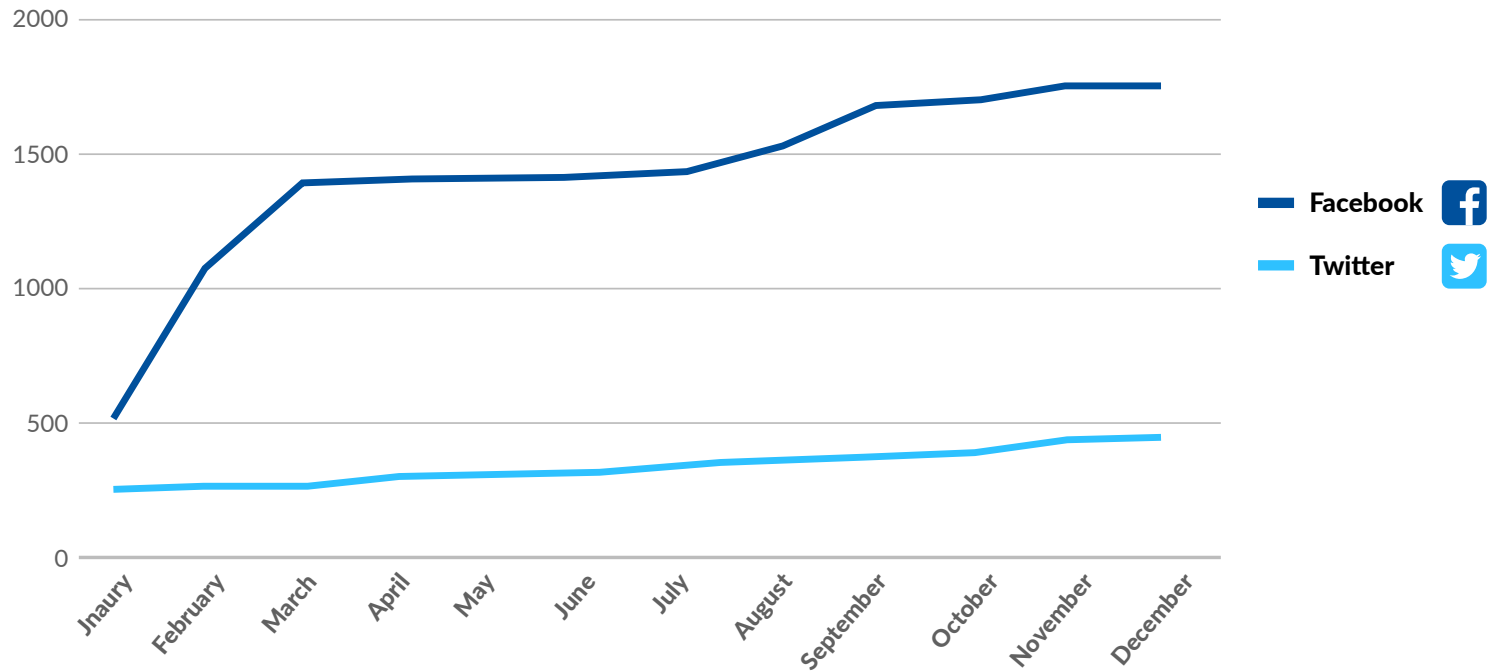


Figure 11. Growth of Facebook and Twitter audiences in 2018.

The Corporation's following on Facebook showed two peaks of reach during 2018. The first was registered in February, with 45.904 people and the second in September, with 8.400 people. CEMarin's following is

desegregated as follows: 65% women, 34% men with the highest concentration of followers being people aged 25 to 34 years old (see Fig. 12). On Twitter, the Corporation reaches about 500 followers, 65% of which are men.

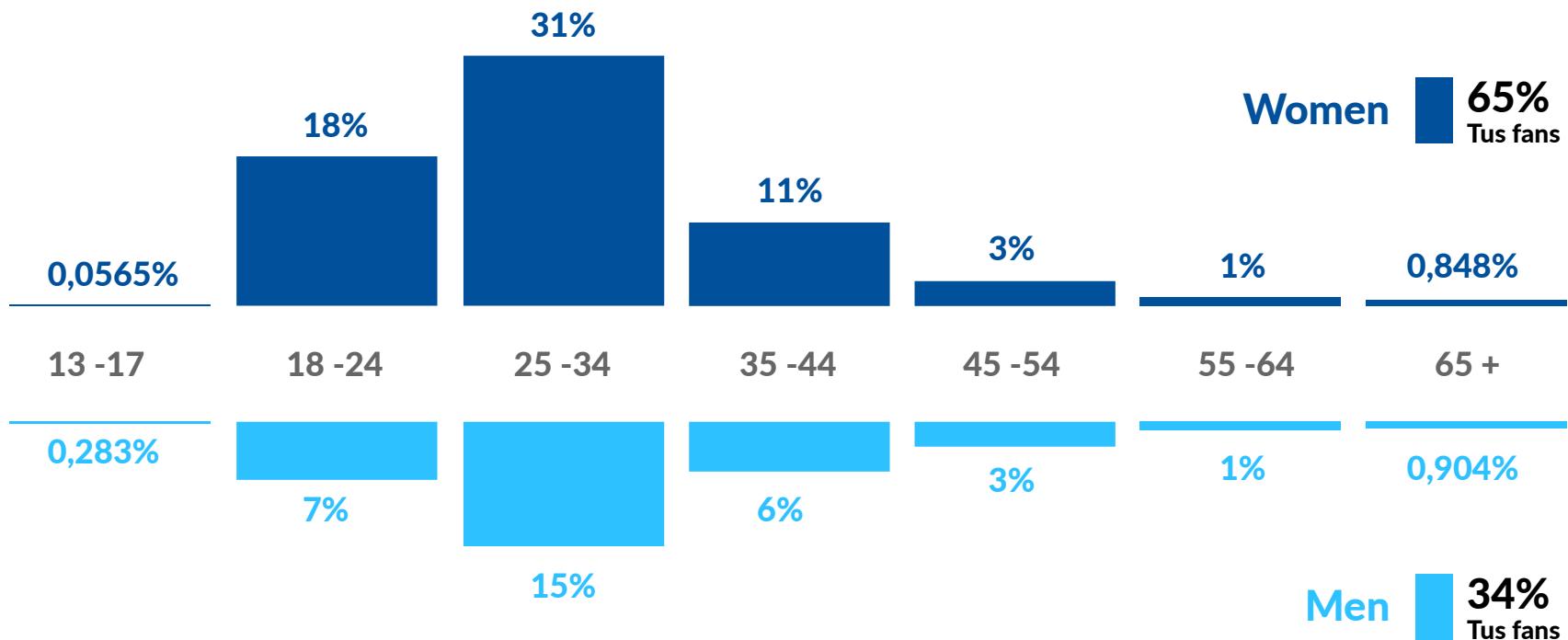


Figure 12. Demography of the audience on Facebook.

During the second semester of 2018 the Corporation's efforts focused on the promotion of the International Conference on Marine Sciences. The distribution of digital content through social networks was mainly done through Facebook with posts that included information about the researches who would participate in the Conference (see Fig. 13). The advertising agency El Grupo

directed and supported the distribution of this content with a digital marketing strategy. Thanks to the strategic alliances established by CEMarin, the event's dissemination was also supported by the Universidad Nacional de Colombia's Press Agency, and the Communication System and the Digital Communication Office of the Universidad de Antioquia.



Javier López Lara



Dr. Javier López Lara is currently the Head of the Coastal Hydrodynamics and Infrastructure Group at IH Cantabria. In 1998, he graduated in Civil Engineer with Major in Structures and, in 2002 he got his Ph.D. in Civil Engineering at University of Cantabria (Spain). His research interest are Focus on numerical modelling using CFD techniques of water wave mechanics, wave-structure interaction, wave generated turbulence, surf-fo hydrodynamics and environmental flows*

*<https://www.linkedin.com/in/javier-lopez-lara-94591952/>



SUPPORT:

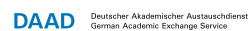


Figure 13. Graphic piece designed for the International Conference on Marine Science.

Google for non-profit organizations

In the second semester of 2018 CEMarin applied to the Google program for non-profit organizations. The program provides tools to increase the visibility of organizations and improve efficiency of their internal processes.

Thanks to this program, as of November 2018, CEMarin Corporation receives 10,000 USD per month in advertising in-kind within the Google Ad Grants search engine marketing platform. Among the main benefits of this tool are the following:

- Google Ad Grants shows the search ads of the Corporation to people who are seeking non-profits entities, facilitating that CEMarin is made known among people who previously did not know it.
- With Google Ad Grants the Corporation can recruit volunteers, attract donations and share its mission with audiences around the world.
- Through Google Ad Grants it is also possible to help partner universities to disseminate their research, events, calls or courses related to marine sciences.
- Lastly, through the Google Ad Grants platform the Corporation can create campaigns about its history, events, courses, calls or whatever it wants to share.

An additional advantage of the Google program for non-profit organizations is the Google Suite, which offers all Google apps for free with the @cemarin.org domain attached: Gmail, Documents, Calendar, Drive or Hangouts. Thanks to this, the CEMarin Corporation was able to change the email accounts to the Google Suite, facilitating collaborative work and keeping all the information in one place.

Google for Nonprofits

Google for Nonprofits is a technology designed for NGO with free access to various products and services.

How did CEMarin get the benefit of Google for Nonprofits?

The administrative area of CEMarin applied to the program with all documents required by Google.



Google Ad Grants

In November 2018, CEMarin receive for the first time USD 10,000 per month in advertising within the marketing platform of AdWords search engines through the Google program for organizations nonprofit.

G- suite for Nonprofits

It provides access to Google Suite free, allows the creation of accounts for mail with domain @cemarin.org and facilitates access to other applications such as: Google Drive, Calendar, Documents, and Hangouts Meet.



Major visibility

Google Ad Grants shows our search ads to people who are looking for Non-profit organizations such as CEMarin and they help us reach people who were previously unaware of our cause.



Efficiency and productivity

G-suit allows to have all the information in one place and create emails accounts with our domain to improve teamwork and the efficiency of daily tasks when accessing all the applications offered by Google.

5. Strategic management

Scientific Committee

The Scientific Committee carried out the following activities during 2018:

1. Drafted and approved the terms of reference for CEMarin calls No. 12 and 13.
2. Evaluated the projects submitted to CEMarin calls No. 12 and 13.
3. Structured the CEMarin call No. 14.
4. Provided guidance for the organization of the International Conference on Marine Sciences.
5. Evaluated potential courses offered by the Corporation.5.2. CEMarin Mini Summit 2018.



CEMarin Mini Summit 2018

Since 2016, the Corporation holds an annual meeting called CEMarin Mini Summit. The objective of the summit is to support the Corporation's strategic management and scientific planning through the ideas provided by the members of the Board of Directors, the Scientific Committee and marine sciences researchers of the partner universities that participate in the event.

The 2018 summit was held at Justus-Liebig University, in Giessen, Germany, from March 5th to 7th. During the summit the research action plan of the Corporation was defined, as well as the areas of research that would be given priority and the type of proposals that would be presented to international cooperation entities. Lastly, the guidelines for projects that would be presented to diverse sources of national and international donors were established.



Mini Summit Report

Scientists and members of CEMarin met at the second minisummit at Justus Liebig University Giessen to propose CEMarin strategic plan, activities 2018 /2019 and project proposals

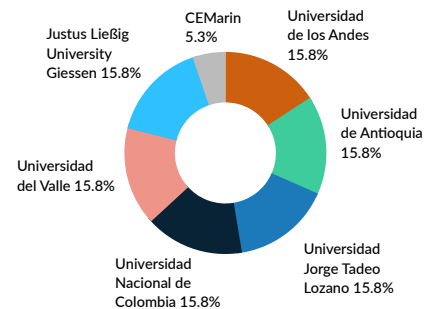


Main objective

Building a guideline for research proposals and planning the applications to national and international funding opportunities on key issues to raise fund.

Activities

- Design CEMarin strategic plan
- Deliberations in working groups
- Summary of working group discussions
- CEMarin activities 2018/19 and project proposals
- DAAD evaluation workshop
- Closing discussions



Three professors of each university attended to the Mini Summit 2018



6. Financial and tax management

Details of the budget execution (income and expenses) during the period from January to December 2018 are presented below.



Income

According to the budget prepared for the year 2018, the income of the Corporation had been estimated at \$ 701.200.000 COP. Due to the additional resources provided by the German Academic Exchange Service - DAAD to cover the expenses of the Interna-

tional Conference on Marine Sciences, CEMarin revenues amounted to \$ 752.339.587 COP and the income of the Corporation exceeded the budgeted amount by 7%.

Concept	Budget	Income	% of participation
Income university fees	\$ 240.000.000	\$ 240.000.000	32%
Fundraising	\$ 300.000.000	\$ 343.734.174	46%
Agreement with the DAAD for salaries	\$ 151.200.000	\$ 151.192.209	20%
Bank interest	\$ 10.000.000	\$ 17.413.204	2%
Bank interest	\$ 701.200.000	\$ 752.339.587	107%

Table 6. Income of CEMarin in 2018.

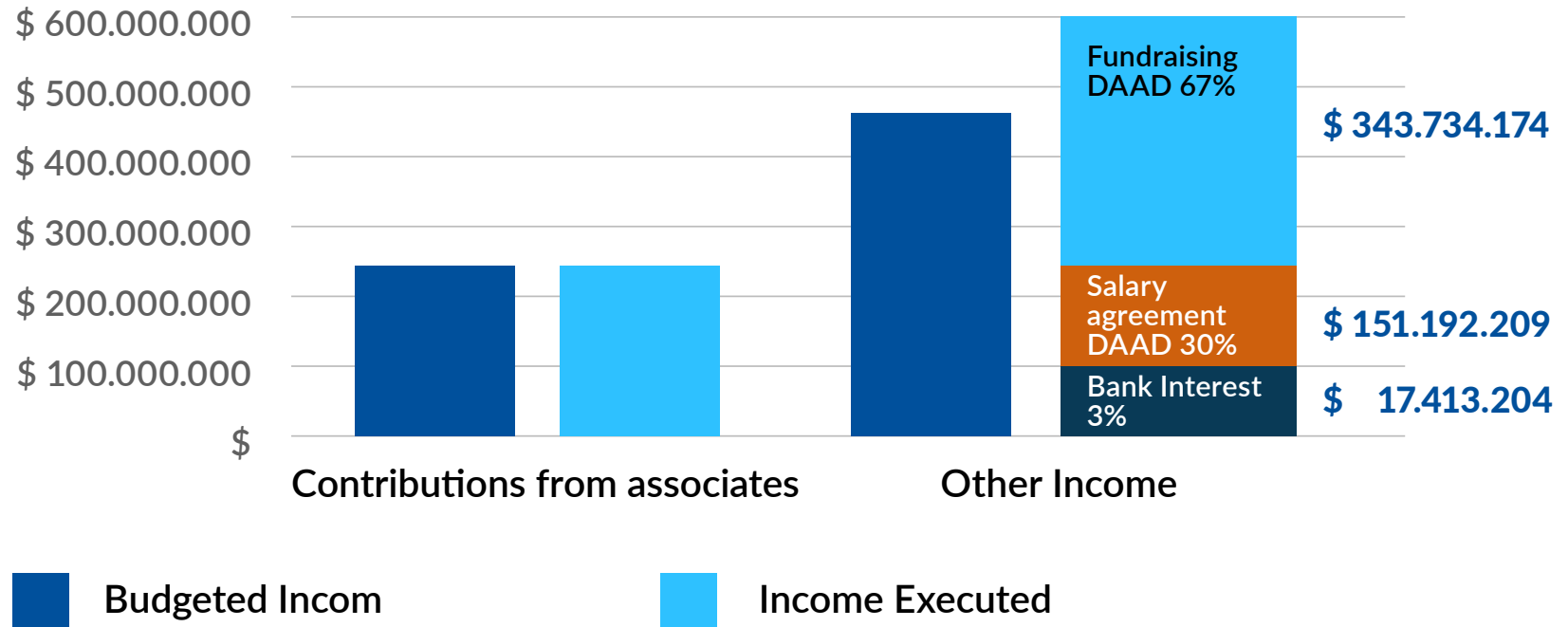


Figure 14. Income of CEMarin in 2018.

The details for the income received by the CEMarin Corporation and provided by the German Academic Exchange Service (DAAD) are shown in Figure 15.

**Total resources
received from
the DAAD:
\$ 484.721.538**

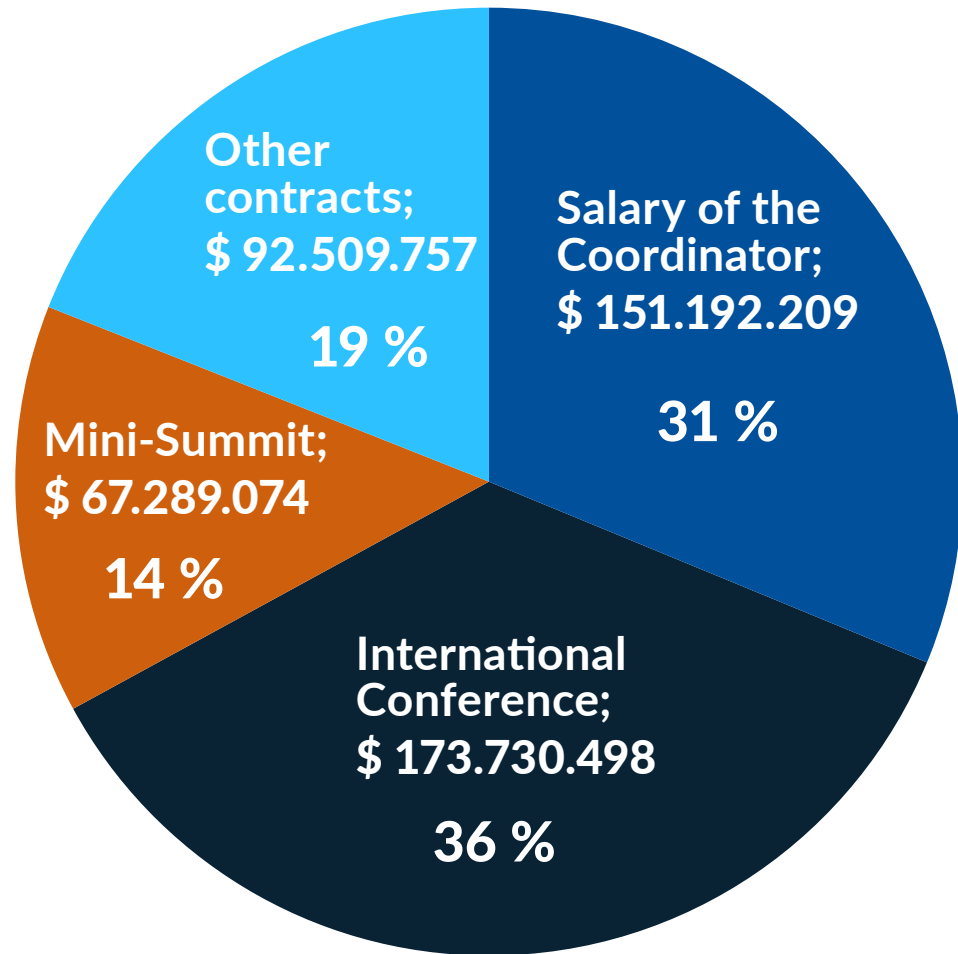


Figure 15. Income received from the DAAD for 2018.

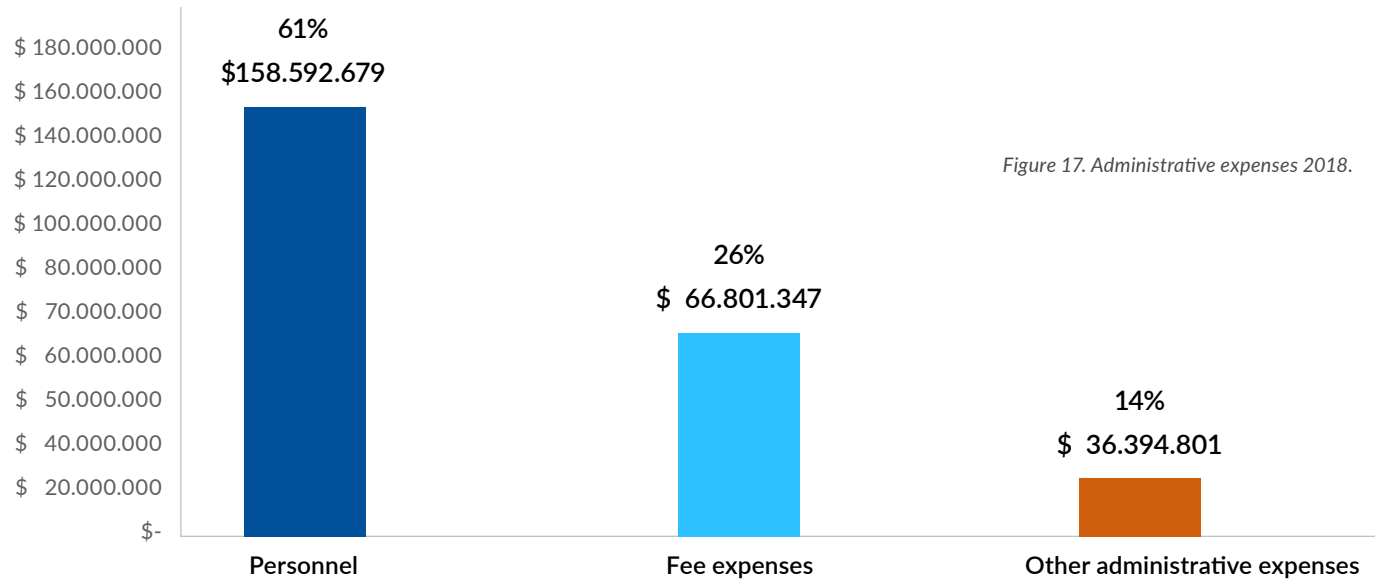
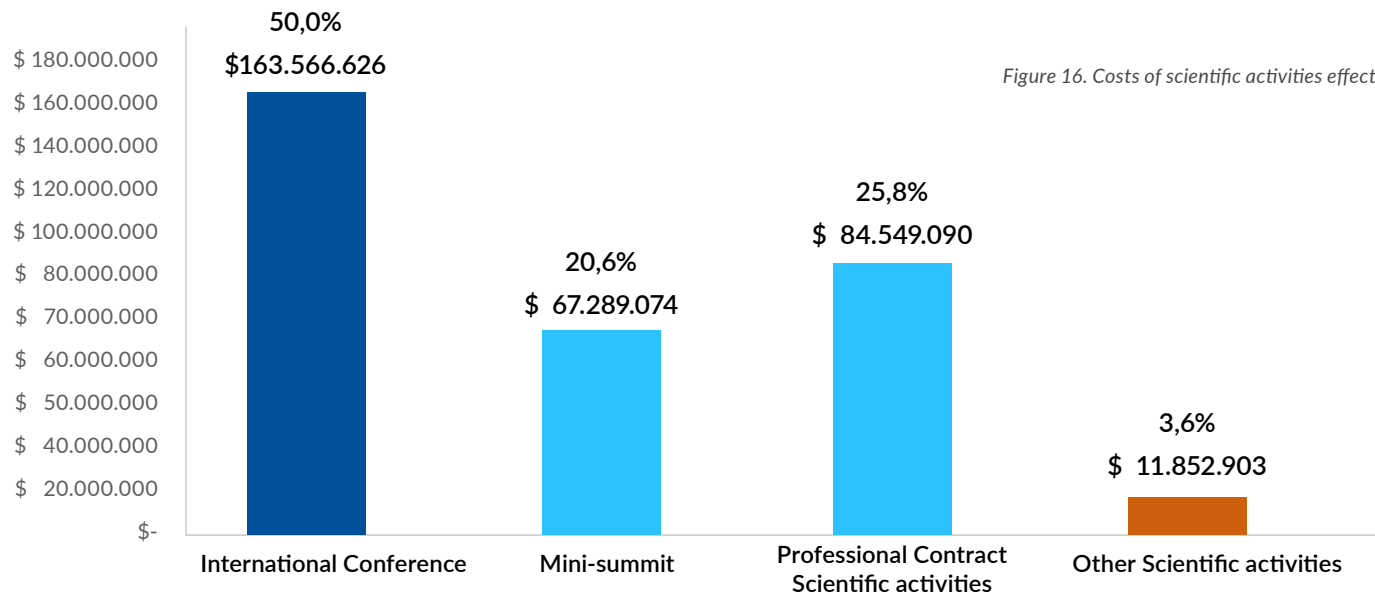
Scientific activities cost

Of the scientific activities programmed for 2018, \$327.257.693 COP were executed, corresponding to 55% of the budget established for this line-item. Of the funds executed, the payment of the scientific

professional fee and the execution of fundraising projects such as the CEMarin Mini Summit and the International Conference on Marine Sciences (\$ 230.855.700 COP), stand out.

Scientific activities programmed for 2018			
Administrative expenses	Budget	Executed	%
Fundraising execution 2018-2019	\$ 300.000.000	\$ 230.855.700	77%
Academic activities	\$ 86.561.297	\$ 2.559.828	3%
Scientific professional fee	\$ 60.000.000	\$ 50.000.000	83%
Communication fees	\$ 27.600.000	\$ 25.326.000	92%
Support for calls	\$ -	\$ 9.223.090	
Travels public relations	\$ 10.000.000	\$ 9.293.075	89%
Promotions, material and publications	\$ 5.000.000	\$ 4.915.000	98%
Support tender calls	\$ 5.000.000	\$ -	0%
Call No.13	\$ 120.000.000	\$ -	0%
Seed capital	\$ 20.000.000	\$ -	0%
Call No.12 Modality 2	\$ 100.000.000	\$ -	0%
Call No.12 Modality 1	\$ 60.000.000	\$ -	0%
Total scientific activities programmed	\$ 794.161.297	\$ 327.257.693	55%

Table 7. Execution of resources in scientific activities effective 2018.



The total cost of scientific activities and administrative expenses for the 2018 period are shown in Figure 18.

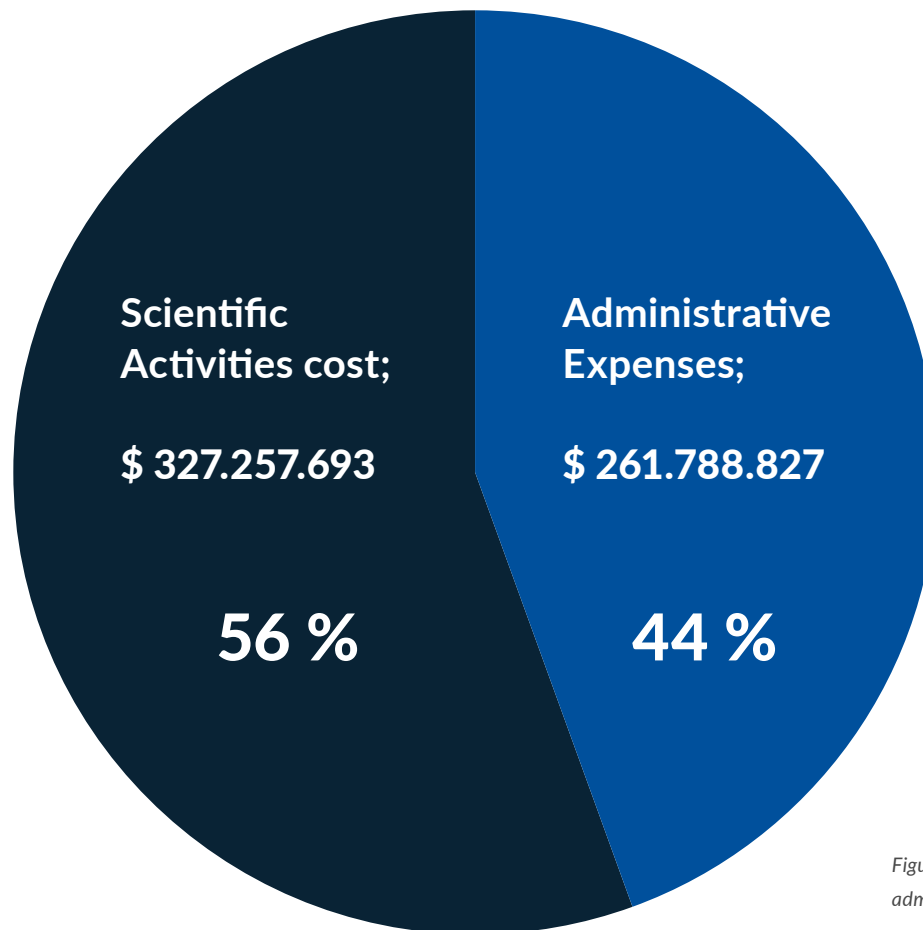


Figure 18. Scientific activities cost and administrative expenses effective in 2018.

Reserve funds and calls in execution

The execution of the calls approved and balance reserve funds is shown in Table 8.

Call	Execution 2018	Pending execution 2019	Balance to be released	Year limit	Observation
No. 5	\$ 77.000.000	\$ 20.000.000	\$ 7.945.850	2019	The balance to be released corresponds to the contract of Nils Hommernick
No. 8	\$ 78.000.000	\$ 24.740.000	\$ 9.025.910	2019	The balance to be released corresponds to balances according to the amount approved in the contracts.
Total	\$ 155.000.000	\$ 44.740.000	\$ 16.971.760		

Table 8. Execution of the Calls No. 5 & 8.

The balance of \$20 million COP to be executed in Call No. 5 corresponds to money that has not yet been disbursed to the CEMarin Early Stage Researchers Juan Guillermo García Garay and Elena Quintanilla Alcaide.

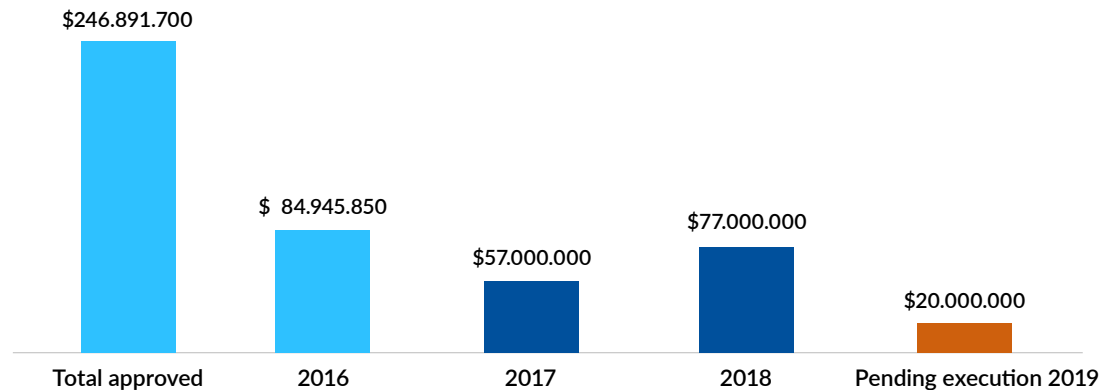


Figure 19. Execution of the Call No. 5.

The balance of \$24,740,000 COP to be executed in Call No. 8 corresponds to money that has not yet been disbursed to CEMarin Early Stage Researchers Argiro de Jesús Ramírez, Juan Carlos Narváez and Gilberto Junior Orozco.

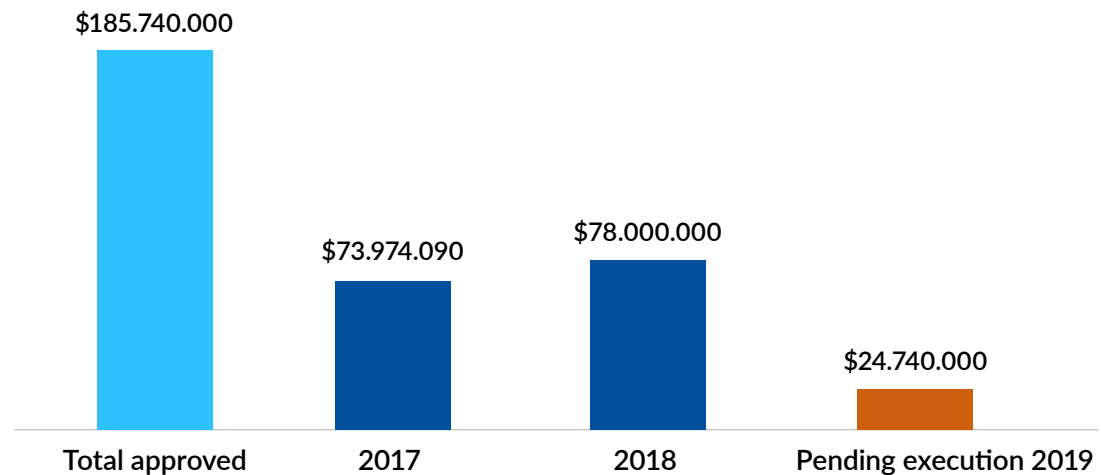


Figure 20. Execution of the Call No. 8.

Resources received from German Academic Exchange Service

The following is a historical summary of the financial resources received by the German Academic Exchange Service (DAAD) to-date.

In 2016, DAAD refunded expenses for the International Workshop on Omic Techniques. In 2017 the resources received from DAAD were used to cover the salary of the Agreements Coordinator and

for the execution of other contracts. Likewise, in 2018 the funds were also used to cover the salary of the Agreements Coordinator and the execution of other contracts, including the salary of the Scientific Project Manager. Additionally, resources were used to develop the Mini Summit in Germany and to carry out the International Conference.

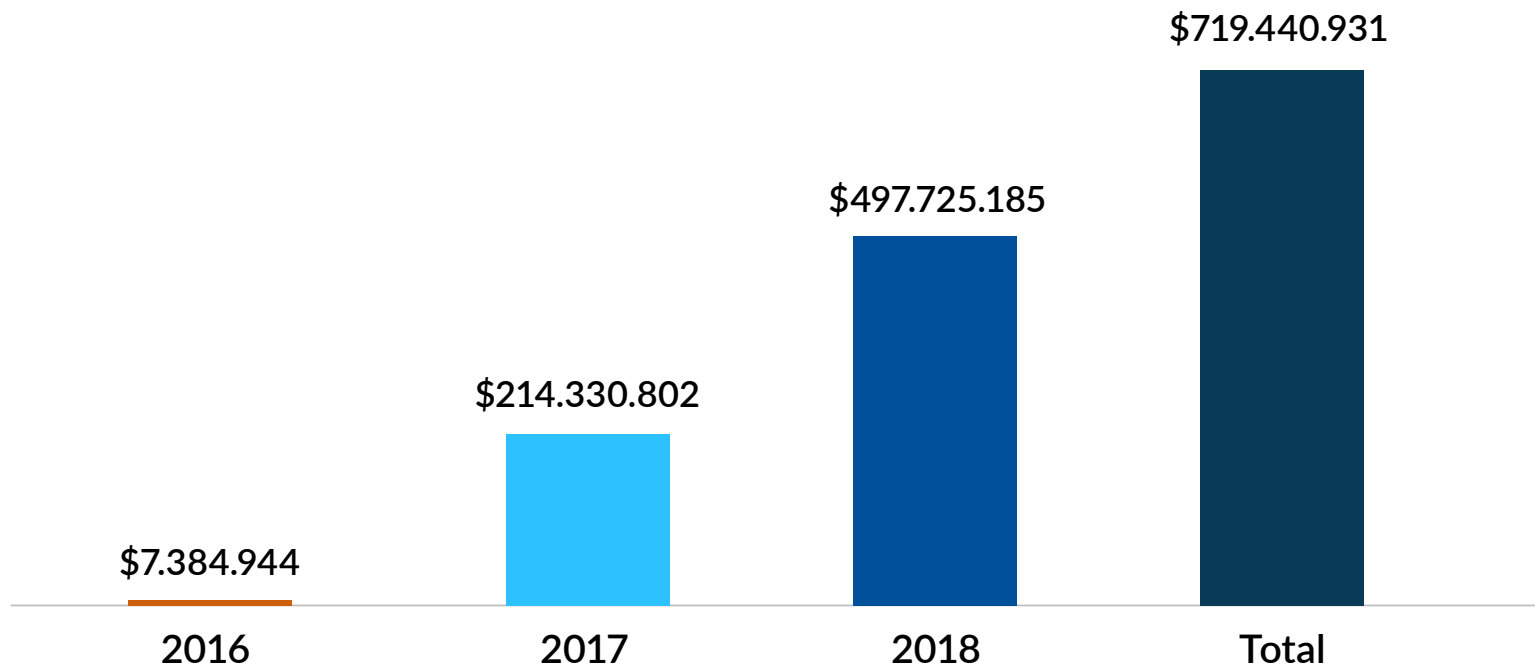


Figure 21. Resources received from the DAAD since the creation of CEMarin.

Tax aspects

With the finance law that came into effect on January 1st, 2019 (Law 1943 of December 28th, 2018) no changes were generated for non-profit entities (ESAL). Consequently, in accordance with the procedure established

by the competent authorities, the CEMarin Corporation must update its web register with the National Direction for Customs and Taxes (DIAN) to continue under the “Special Tax Regime”.

Permanence	It corresponds to the process that must be advanced for a single time, the entities referred to in transitory paragraphs 1 and 2 of article 19 of the tax statute.
Qualification	It corresponds to the process to be advanced by the nonprofit entities referred to in subsection 1 of article 19 of the tax statute, which aspire to be taxpayers of the special tax regime of income tax and complementary and entities that were excluded or renounced special tax regime of income tax and complementary, and want to opt again to belong to this regime.
Upgrade	Corresponds to the mandatory process subsequent to the qualification or permanence that must be advanced annually by the taxpayers referred to in this section, who decide to stay within the special tax regime of income and supplementary tax.

Figure 22. Process to belong to the Special Tax Regime.

According to article 356-3 of the Tax Statute of Colombian legislation, in 2019 CEMarin upload to the DIAN website the Corporation’s administrative and financial Documentation required within the deadlines established by the entity.

An underwater photograph with a blue-green tint. A diver is visible in the center, surrounded by a massive, dense school of small fish that fills much of the left and center of the frame. The background shows some dark, rocky structures and more distant fish.

7. Human resources

During 2018 the CEMarin administrative staff was comprised of an interdisciplinary team of professionals who were an indispensable part of the successful execution of all the activities that were developed throughout the course of the year.



CEMarin staff members 2018



Jairo Zapata Martínez
Executive director



Marcela Sánchez
Networks coordinator



Mauricio Romero
Manager, Scientific Projects



Doris Tovar
Public accountant



Angélica Santacruz
Manager of Scientific Calls



Aída Acevedo
Administrative assistant



Carina Marek
Alumni coordinator



Elizabeth Reina
Statutory Auditor



Lina Marcela Gallo
Social communicator



Jorge Mario Ordóñez
Graphic Designer

Figure 23. 2018 CEMarin Administrative staff.

Some of the main novelties in the configuration of the work team that are worth mentioning, are highlighted below:

In July a call was opened to hire a new Administrative Assistant and a professional who studied business administration was selected for the position. The professional that until then had occupied that job was made the new Calls Manager in September of the same year.

A communications specialist was hired to support the International Conference on Marine Sciences and lead the logistics in Medellín. A graphic designer was also hired to develop the communication strategy and the design of the advertising products.

The Coordinator of Agreements, Manager of Scientific Projects, Alumni Coordinator, Public Accountant, and the Statutory Auditor all renewed their contracts with the Corporation, maintaining the positions they occupied.

Finally, the Executive Director, Professor Jairo Zapata Martínez, held his position until December 31st, 2018, fulfilling his second and last leadership period, in accordance with the provisions of CEMarin Board of Directors.

2018

Management report

